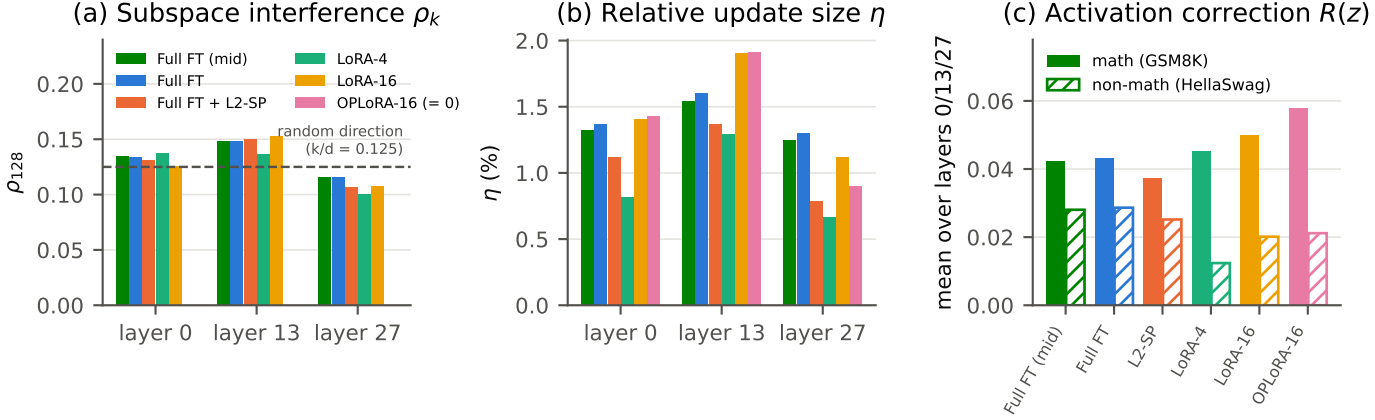
(a) Subspace interference ρ_k
ρ_{128}
0.20
0.15
0.10
0.05
0.00
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16 (= 0)
random direction
(k/d = 0.125)
layer 0
layer 13
layer 27
(b) Relative update size η
η (%)
2.0
1.5
1.0
0.5
0.0
layer 0
layer 13
layer 27
(c) Activation correction $R(z)$
mean over layers 0/13/27
0.06
0.04
0.02
0.00
math (GSM8K)
non-math (HellaSwag)
Full FT (mid)
Full FT
L2-SP
LoRA-4
LoRA-16
OPLoRA-16